Social Learning with Logical Formulas

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We study social learning with logical formulas.

1 Introduction

Does social learning work when opinions are more sophisticated representations of the world, like logical formulas? We find out.

There is work that look at opinion dynamics with logical formulas [Schwind et al., 2015, 2016; Christoff and Grossi, 2017; Botan et al., 2019; Schwind et al., 2024, 2026], but usually with an eye towards consensus. It seems that the truth-tracking dimension on a network has not been tackled yet.

Contributions. We show interesting things.

Related Literature. Belief merging operators [Konieczny and Pino Pérez, 2011] and their truth-tracking properties [Everaere et al., 2010, 2020].

Some precursors in the belief merging literature [Schwind et al., 2015, 2016, 2024, 2026], though these only look at convergence (no truth involved).

The opinion diffusion literature [Grandi et al., 2015; Christoff and Grossi, 2017; Botan et al., 2019; Grandi et al., 2021].

The computational literature [Betz, 2013].

The DeGroot dynamics [DeGroot, 1974], and the wisdom aspect, where we have a ground truth [Golub and Jackson, 2010].

2 Model

Agents. *Agents* are the nodes of a finite, directed graph $G = (N, E)$, where $N = \{1, \dots, n\}$ is the set of agents, and E is the set of edges. We write $N(i)$ for i's out-neighborhood in G, i.e., $N(i) = \{j \mid (i, j) \in E\}$, the set of agents that i has an edge to. Intuitively, $N(i)$ is the set of agents that i pays attention to G.

Opinion Dynamics. There is a set $A = \{a_1, \dots a_m\}$ of *atoms*, with L being the set of *logical formulas* constructed using these atoms using the usual connectives ($\neg$, $\wedge$, $\vee$, $\rightarrow$, $\bot$ and $\top$). We call 2^A the set of *possible worlds*, or *truth-value assignments*. If $\varphi \in L$ is a propositional formula, $[\varphi] \subseteq 2^A$ is the set of models of φ, i.e., the truth-value assignments that make φ true.

The dynamics occurs in rounds $t \in \{0, 1, \dots\}$. For each agent i, $\varphi_i^t \in L$ is agent i's opinion at t. We assume opinions at $t = 0$ are given by a noise model (see below). After

this, agents update their opinions using the opinions of their neighbors and a merging operator M:

$$\varphi_i^{t+1} = M\big(\varphi_j^t \mid j \in N(i)\big),$$

i.e., i's opinion at $t+1$ is the result of merging the opinions at t of i's neighbors in G. Note that i's own opinion at t enters here if we assume there is an edge from i to itself G.

Merging Operators. What merging operators shall we use?

Noise Model. Here we describe a noise model that generates initial opinions.

We assume, first of all, that there exists a ground truth state of the world $w^\star \in 2^A$. [TBC]

Questions. Do we get that each agent's beliefs stabilize, i.e., is there some belief φ_i^* such that:

$$\lim_{t\to\infty} \varphi_i^t = \varphi_i^*,$$

for any agent i? Under what conditions does this happen?

If agents' beliefs stabilize, do we see consensus? In other words, is there some belief φ^* such that $\varphi_i^* = \varphi^*$, for any agent i? And under what conditions?

If there is consensus, do we see consensus to the truth as n grows? Take a family of graphs G_n in which the dynamics reaches consensus $\varphi^*(n)$ as $t \to \infty$. Call a family of graphs G_n *wise* if:

$$\lim_{n\to\infty} \Pr\Big[\big[\varphi^*(n)\big] = w^\star\Big] = 1,$$

i.e., if the consensus belief over the whole graph G_n is exactly (a formula with the model) $w^\star$, the ground truth state, with high probability, as n grows. Our question, then, is what kind of networks are wise, and for which noise models?

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